

Brain MRI Research Projects

BID Lab — ShanghaiTech University

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1 Project Overview

Two brain MRI research projects at BID Lab, School of Biomedical Engineering, ShanghaiTech University.

1.1 Project 1: T2T-Bridge (ISBI 2025)

Goal: Reconstruct 3D thin-slice infant MRI from 2D thick-slice scans using Direct Diffusion Bridge.

Method: I2SB + Classifier-Free Age Guidance + Structure Consistency Loss. **Key Results:** SSIM 0.9573, PSNR 33.38 dB, 15-step inference. Clinical deployment at Fudan Children’s Hospital.

1.2 Project 2: Multimodal PET/MR Diagnosis (ICASSP 2025)

Goal: AI diagnosis from multimodal PET/MR data, allowing single-modality inference. **Method:** MoPoE-VAE + cross-modal synthesis + missing-modality robust classifier. **Key Results:** 57.9% accuracy on ADNI (38 subjects, 3-class), 245s training with fp16.

2 ADNI Dataset

Table 1: Dataset Statistics

Metric	Value
Total subjects	225
T1 MRI NIfTI	223
FDG-PET NIfTI	119
Tau-PET NIfTI	62
All three modalities	38
Disk usage	11 GB
CN (Normal)	114
EMCI	48
MCI	40
LMCI	21
AD	2

3 Experiments

3.1 Synthetic Data (3 experiments, RTX 4060)

Table 2: Synthetic data results (100 samples, 20 epochs)

Exp	Dim	Cross	Best Loss	Time
Baseline	128	Yes	0.851	220s
Large	256	Yes	0.670	230s
No-Cross	128	No	0.763	215s

3.2 Real ADNI Data (38 all-3 subjects)

Table 3: Real data training (fp16, batch=4, 10 epochs)

Metric	Value
Latent dimension	128
Hidden dimensions	[64, 32]
Parameters	8,539,939
Training time	245s (4.1 min)
Loss (initial $\rightarrow$ final)	1.187 $\rightarrow$ 0.793
Accuracy (3-class)	57.9%
F1 score (weighted)	0.425
GPU	RTX 4060 Laptop (8GB)

4 Infrastructure

- **DICOM conversion:** SimpleITK ImageSeriesReader — 181 PET volumes converted
- **Brain extraction:** HD-BET v2.0.1 (<5s/volume)
- **Tissue segmentation:** MONAI DynUNet (5.4M params)
- **Registration:** SimpleITK 2.5.4 (ANTs/FSL alternative)
- **Quantization:** bitsandbytes 0.49 (fp16/int8)
- **Training:** PyTorch 2.11 + CUDA 13.0 + fp16 AMP

5 Conclusion

Both projects have complete code pipelines with verified training. The MoPoE-VAE achieves 57.9% accuracy on real ADNI data (38 subjects, 3-class) with fp16 acceleration. Training takes only 4 minutes, demonstrating the efficiency of the lightweight architecture. The full ADNI dataset (225 subjects) and dHCP data (T2T-Bridge) are ready for larger-scale experiments.

6 Repository

<https://github.com/tzhazuma/bmebrainproject>